

SEHATI - EMR UKS Sekolah Rakyat

An integrated digital healthcare system for school clinic documentation, student health records, medication inventory, reporting, and nursing decision support.

EMR UKS

FastAPI

SQLite/PostgreSQL

Role-Based Access

Project Overview

SEHATI is an Electronic Medical Record (EMR) application designed for school health unit services. It helps healthcare staff document student visits, manage patient records, record medication use, generate reports, and maintain a more organized and traceable clinical administration workflow.

50+

API endpoints

2

User roles

PDF/XLSX

Report outputs

NANDA

Care support

Clinical & Operational Features

- Student/patient identity records and school clinic visit history.
- Documentation of complaints, examination, diagnosis, treatment, notes, referral, and follow-up.
- Medication administration records per visit and minimum stock monitoring.
- Daily and monthly visit reports, PDF/Excel export, and medication mutation reports.

Governance & Safety

- Login with JWT authentication and signed session cookies.
- Admin and nurse roles to restrict page and endpoint access.
- Audit logs for key actions such as login, data edits, and user management.
- Local database backup and migration support to PostgreSQL on Railway.

Role Contribution

- Translated school clinic service needs into a digital workflow covering student registration, visits, treatment, medication, referrals, and reports.
- Provided a nursing and healthcare administration perspective so terminology, forms, and report outputs match real clinical workflows.
- Promoted structured, searchable, and accountable documentation for school healthcare service activities.

SEHATI Application Preview

A visual preview of the EMR application interface used as supporting project evidence.



SEHATI login page - an integrated digital healthcare service system for Sekolah Rakyat.

Technical Scope

- Backend API built with FastAPI, using local SQLite storage and PostgreSQL support for deployment.
- Data model includes users, audit logs, patients, assessments, recommendations, school clinic visits, medication records, inventory, CKG events, screening stations, referrals, and recommendation letters.
- Deployment prepared with Dockerfile, Procfile, railway.json, Railway documentation, and environment variable configuration.
- Additional integrations include AI care recommendations, WhatsApp notifications via Fonnte, and PDF/Excel report exports.

Portfolio Narrative

This project demonstrates the ability to combine nursing experience, healthcare administration, and digital workflow understanding. Its core value is improving school clinic documentation, reporting, and service traceability while preserving healthcare professionals' clinical judgment.